

Exp 4

Aim

To create a cryptocurrency account using **MetaMask** and transfer a specified amount of cryptocurrency from one wallet account to another using the **Sepolia test network**.

Theory

MetaMask is a cryptocurrency wallet that allows users to create and manage blockchain accounts and interact with decentralized applications. Each account has a unique **public wallet address** and a **private key** used to digitally sign transactions.

For this experiment, the **Sepolia Ethereum test network** is used. Sepolia ETH is test cryptocurrency and has no real monetary value, making it suitable for educational purposes.

A cryptocurrency transfer involves:

1. Selecting the sender's wallet.
2. Providing the receiver's wallet address.
3. Specifying the amount of cryptocurrency to transfer.
4. Creating and signing the transaction using the sender's private key.
5. Broadcasting the transaction to the blockchain network.
6. Waiting for the transaction to be confirmed.

A small **gas fee** is required to process an Ethereum transaction.

Install the required Python library using:

```
pip install web3
```

Procedure

Step 1: Create MetaMask Account

1. Install the MetaMask browser extension.
2. Open MetaMask.
3. Select **Create a New Wallet**.
4. Set a wallet password.
5. Securely save the Secret Recovery Phrase.
6. Complete the phrase verification.
7. The MetaMask wallet is now created.

Step 2: Create Second Account

1. Open the account selector in MetaMask.
2. Select **Add account**.
3. Create a second account.
4. Note the wallet address of the second account.

The two accounts are:

Account 1 → Sender

Account 2 → Receiver

Step 3: Use Sepolia Test Network

1. Select the **Sepolia** network in MetaMask.
2. Obtain a small amount of Sepolia test ETH from a testnet faucet.
3. Ensure that Account 1 has sufficient test ETH to perform the transaction and pay the gas fee.

Step 4: Execute the Python Program

Run the transfer program:

```
python transfer.py
```

The program asks for:

- Sender's private key
- Receiver's wallet address
- Amount of ETH to transfer

For example:

Enter Sender Private Key:

Enter Receiver Wallet Address:

Enter amount of ETH to transfer: 0.001

The program then creates, signs, and broadcasts the transaction to the Sepolia network.

Step 5: Confirm Transaction

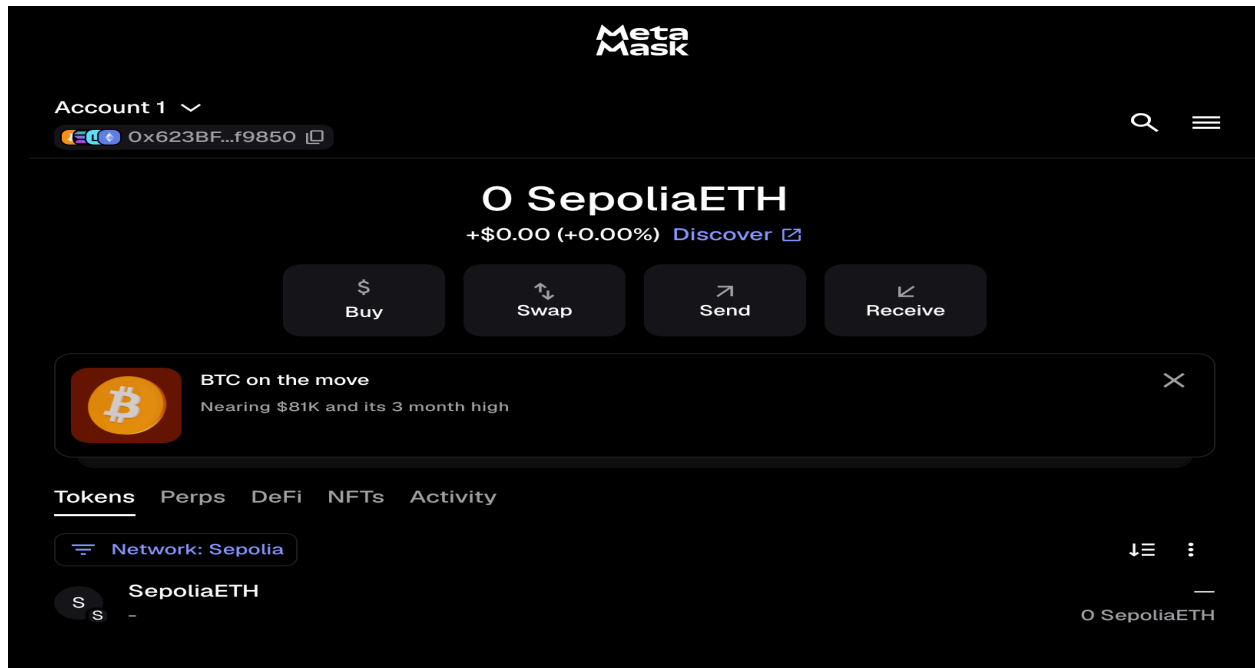
After confirmation, the program displays the transaction hash.

The transaction can be verified using the Sepolia blockchain explorer.

Output:

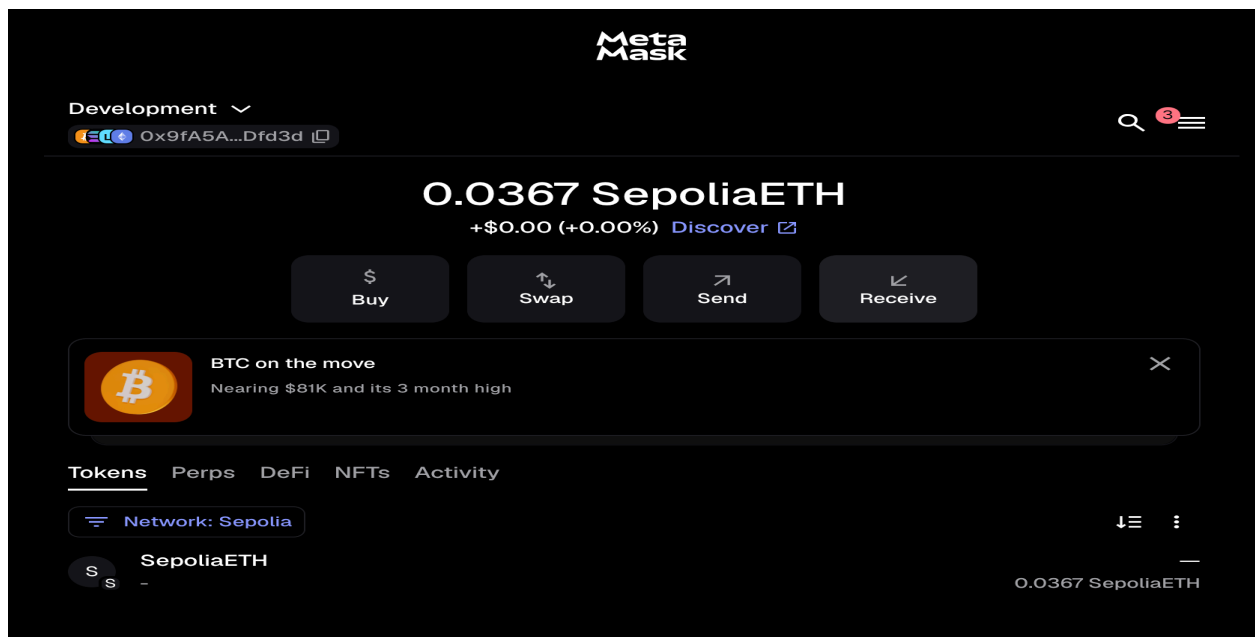
Screenshot 1 — MetaMask Wallet Created

Show the MetaMask wallet dashboard after successfully creating the account.



Screenshot 2 — Sepolia Network and Balance

Show Account 1 selected with the **Sepolia** network and available test ETH.



Screenshot 3 — Successful Transaction

Show the program output after the transaction is successfully submitted and confirmed.

```
(.venv) [chirag@arch BC_Exp]$ python exp4.py
✅ Connected to Sepolia network
Chain ID: 11155111
Enter Sender Private Key:

Sender Wallet:
0x9fA5A4c3386e333e520883b1649A0bca99BDfd3d

Enter Receiver Wallet Address: 0x623BFb0fbCf5ACa22b8FCdc2C4fb2a27c7bf9850

Sender Balance: 0.036691303842534523 ETH

Enter amount of ETH to transfer: 0.0005

Gas Price: 1.000515166 Gwei
Estimated Gas: 0.000021010818486 ETH

Transaction Details
-----
From : 0x9fA5A4c3386e333e520883b1649A0bca99BDfd3d
To   : 0x623BFb0fbCf5ACa22b8FCdc2C4fb2a27c7bf9850
Amount: 0.0005 ETH

Confirm transaction? (yes/no): yes

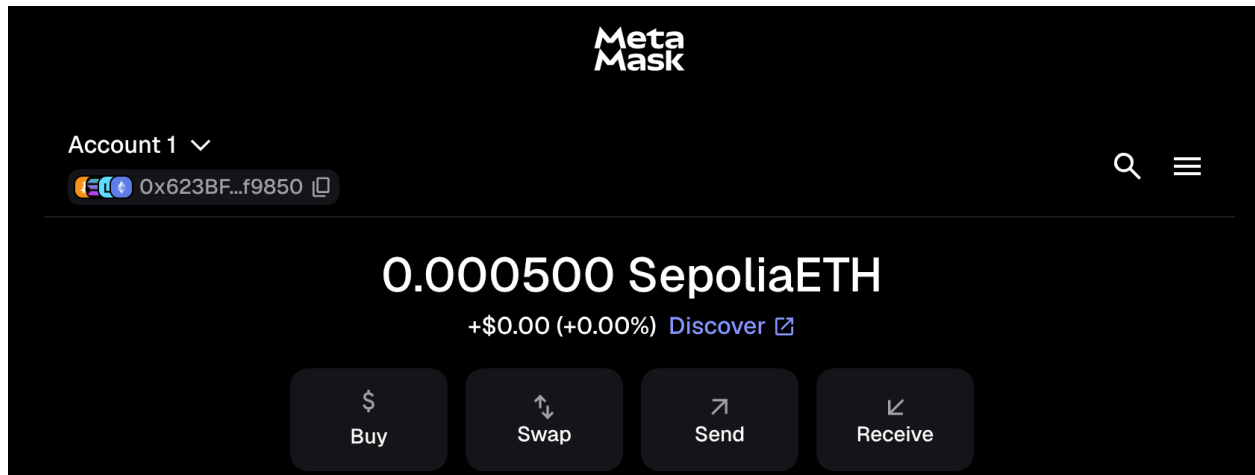
🚀 Transaction sent!
Transaction Hash:
0x1b8496c30c150c03486ce7abb1890be01a15a30ea9bc47c926a30ad90a640d9e

Waiting for confirmation...

✅ Transaction successful!
```

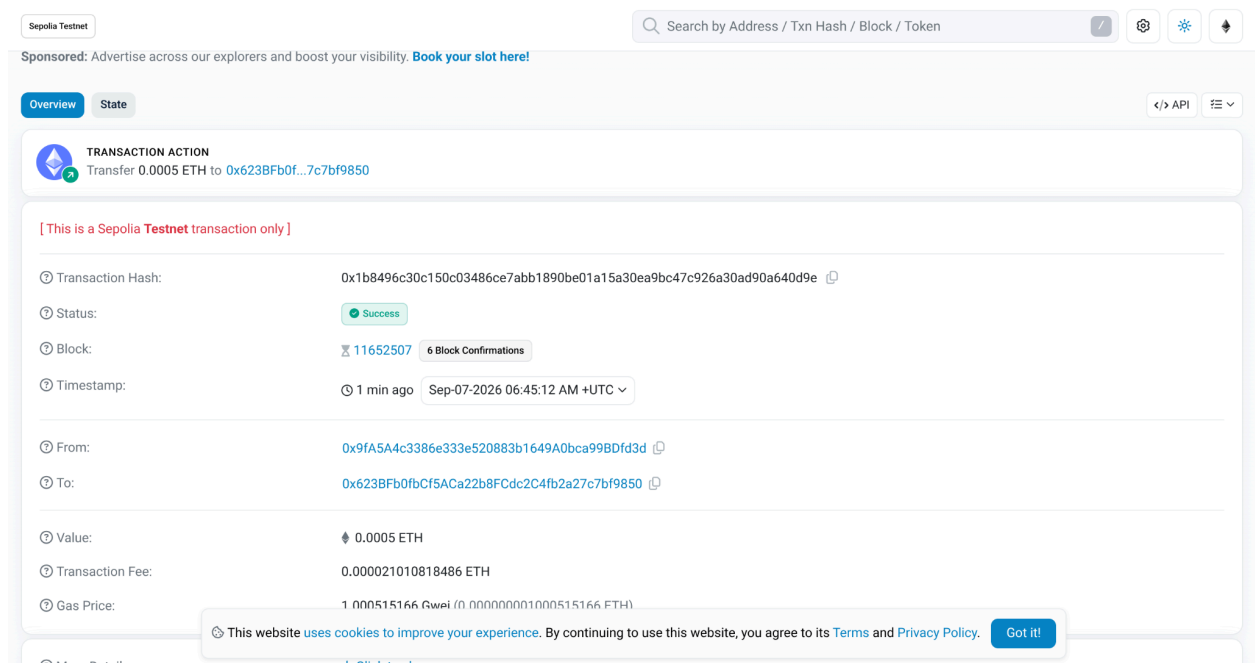
Screenshot 4 — Receiver Account

Open Account 2 in MetaMask and show the updated balance.



Screenshot 5 — Blockchain Transaction Verification

Open the transaction hash in the Sepolia blockchain explorer and show the transaction status.



Conclusion

Thus, a MetaMask cryptocurrency wallet was successfully created and a cryptocurrency transfer was performed between two accounts using the Sepolia test network. The transaction was successfully signed, broadcast, confirmed, and verified on the blockchain.